



# NORTH SYDNEY BOYS HIGH SCHOOL

## 2016 PRELIMINARY ASSESSMENT TASK 2

# Mathematics

### General Instructions

- Working time – 50 minutes
- Write on the lined paper in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Marks may not be awarded for carelessly or badly arranged work.
- Attempt all questions

**Class Teacher:**

☐ Mr Zuber

Student Number: \_\_\_\_\_

(To be used by the exam markers only.)

| Question No | 1               | 2             | 3               | 4             | Total           | Total %           |
|-------------|-----------------|---------------|-----------------|---------------|-----------------|-------------------|
| Mark        | $\frac{13}{13}$ | $\frac{7}{7}$ | $\frac{13}{13}$ | $\frac{6}{6}$ | $\frac{39}{39}$ | $\frac{100}{100}$ |

**Question 1 (13 marks)**

**Marks**

(a) Find the value of  $\theta$ , correct to the nearest minute, if  $\cot \theta = \frac{5}{3}$ , for  $0^\circ \leq \theta \leq 360^\circ$ .

1

(b) If  $\tan \theta = \frac{5}{11}$  and  $\cos \theta$  is negative, find the exact value of  $\sec \theta$ .

2

(c) Find the exact value of

i)  $3 \tan 300^\circ$

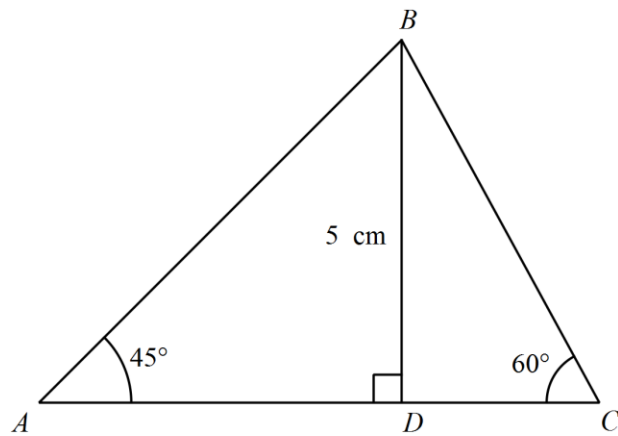
1

ii)  $\frac{2 - \cos 60^\circ}{\sec^2 135^\circ}$

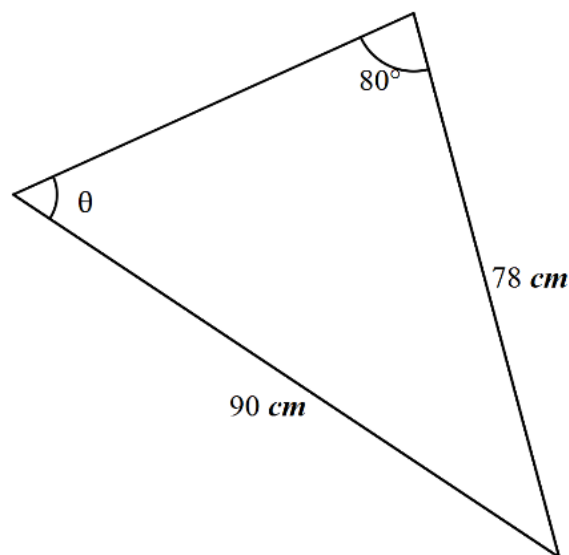
3

(d) Find the exact length of  $AC$ .

2



(e)



i) Find the value of  $\theta$ , correct to nearest minute.

2

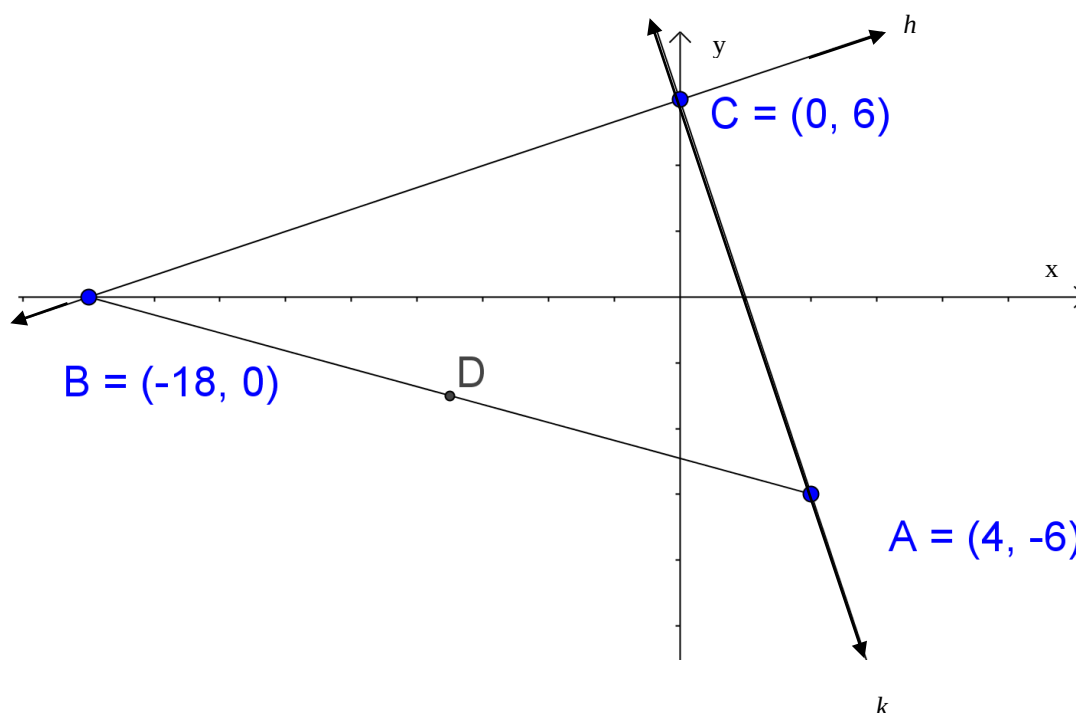
ii) Find the area of the triangle, to the nearest  $\text{cm}^2$ .

2

**Question 2 (7 marks)**

**Marks**

In the diagram, the lines  $h$  and  $k$  are drawn. The coordinates of  $A, B$  and  $C$  are  $(4, -6)$ ,  $(-18, 0)$  &  $(0, 6)$  respectively.  $D$  is the midpoint of  $AB$ .



- |     |                                                                                                                                                                  |   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| (a) | Find the coordinate of $D$ .                                                                                                                                     | 1 |
| (b) | Find the length of $AB$ in exact form.                                                                                                                           | 1 |
| (c) | Find the equation of the line $AB$ in general form.                                                                                                              | 2 |
| (d) | Find the perpendicular distance of $AB$ from the point $C$ , using the perpendicular distance formula $d_{\perp} = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ . | 2 |
| (e) | Hence, find the area of $\triangle ABC$ .                                                                                                                        | 1 |

**Question 3 (13 marks)**

**Marks**

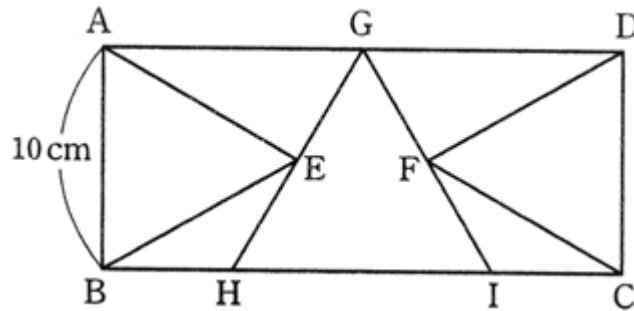
- |     |                                                                       |   |
|-----|-----------------------------------------------------------------------|---|
| (a) | Find all values of $x$ with $0^\circ \leq x \leq 360^\circ$ for which |   |
|     | i) $3 + 2 \cos x = 5 \cos x$                                          | 2 |
|     | ii) $\operatorname{cosec}^2 x = 2$                                    | 3 |
|     | iii) $2 \sin 2x = \sqrt{3}$                                           | 4 |
| (b) | Prove that $(1 - \tan x)^2 + (1 + \tan x)^2 = 2 \sec^2 x$ .           | 3 |

**Question 4 (6 marks)**

(a) Sketch the curve  $y = 4 \sin \frac{x}{2} - 1$ , for  $0^\circ \leq x \leq 360^\circ$

**3**

(b) The rectangle ABCD with one side of 10cm, has three equilateral triangles inside as below.  $\triangle ABE \equiv \triangle CDF$ , and  $\triangle BHI$  is an equilateral triangle but not congruent with other two triangles.



i) Prove that  $\angle AEG = 90^\circ$

**1**

ii) Hence, find the length of AD.

**2**

- End of examination -

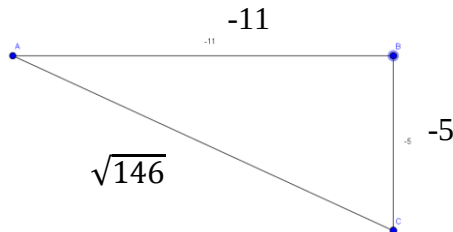
## 2016 PRELIMINARY ASSESSMENT TASK 2 – suggested solutions

### Question 1

(a)  $\frac{1}{\tan \theta} = \frac{5}{3}$

$$\tan \theta = \frac{3}{5}, \quad \theta = \tan^{-1}\left(\frac{3}{5}\right) \\ = 30^\circ 58', 210^\circ 58' \quad \checkmark$$

(b)



$$\sec \theta = \frac{\text{hyp}}{\text{adj}} = \frac{\sqrt{146}}{-11} = -\frac{\sqrt{146}}{11} \quad \checkmark \checkmark$$

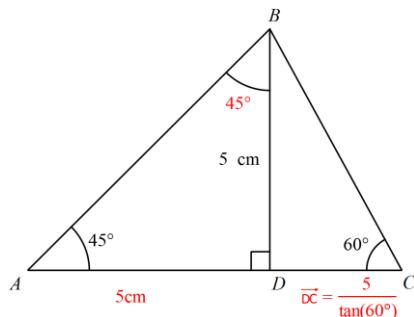
(c)

i)  $3 \tan 300^\circ = 3 \tan(-60^\circ)$   
 $= 3 \times -\sqrt{3} = -3\sqrt{3} \quad \checkmark$

ii)  $\frac{2 - \frac{1}{2}}{\frac{1}{\cos^2(180-45)}} = \frac{\frac{3}{2}}{\frac{1}{\left(-\frac{1}{\sqrt{2}}\right)^2}}$

$$= \frac{1}{2} \times \frac{\frac{3}{2}}{1} = \frac{3}{4} \quad \checkmark \checkmark \checkmark$$

(d)



$$AC = 5 + \frac{5}{\tan 60} = 5 + \frac{5}{\sqrt{3}} \text{ or } \frac{15 + 5\sqrt{3}}{3}$$

✓✓

(e)

i)  $\frac{\sin \theta}{78} = \frac{\sin 80^\circ}{90}$

$$\sin \theta = \frac{78 \times \sin 80^\circ}{90}$$

$$\theta = 58^\circ 36' \text{ or } 121^\circ 24'$$

But  $121^\circ 24'$  is not possible as  $121+80$  is over  $180^\circ$  for angle sum of triangle! ✓✓

ii)  $A = \frac{1}{2} \times 90 \times 78 \times \sin 41^\circ 24'$   
 $= 2321 \text{ cm}^2 \quad \checkmark \checkmark$

### Question 2

(a)  $D = (-7, -3) \quad \checkmark$

(b)  $AB = \sqrt{(4 - (-18))^2 + (-6 - (0))^2}$   
 $= \sqrt{520} \text{ units} \quad \checkmark$

(c)  $3x + 11y + 54 = 0 \quad \checkmark \checkmark$

(d) 10.52 units (2 d.p) ✓✓

(e)  $239.89 \text{ units}^2$  (2 d.p) ✓

### Question 3

(a)

i)  $3 = 3 \cos x, \quad 1 = \cos x$   
 $x = 0^\circ, 360^\circ \quad \checkmark \checkmark$

ii)  $\frac{1}{\sin^2 x} = 2$   
 $\sin^2 x = \frac{1}{2}, \quad \sin x = \pm \frac{1}{\sqrt{2}}$

$$x = 45^\circ, 135^\circ, 225^\circ, 315^\circ \quad \checkmark \checkmark \checkmark$$

iii)  $\sin 2x = \frac{\sqrt{3}}{2} \quad 0^\circ \leq 2x \leq 720^\circ$

$$2x = 60^\circ, 120^\circ, 420^\circ, 480^\circ$$

$$\therefore x = 30^\circ, 60^\circ, 210^\circ, 240^\circ \quad \checkmark \checkmark \checkmark \checkmark$$

## 2016 PRELIMINARY ASSESSMENT TASK 2 – suggested solutions

### Question 3 continued

(b)  $(1 - \tan x)^2 + (1 + \tan x)^2 = 2 \sec^2 x$

$LHS = (1 - \tan x)^2 + (1 + \tan x)^2$

$= 1 - 2 \tan x + \tan^2 x + 1 + 2 \tan x + \tan^2 x$

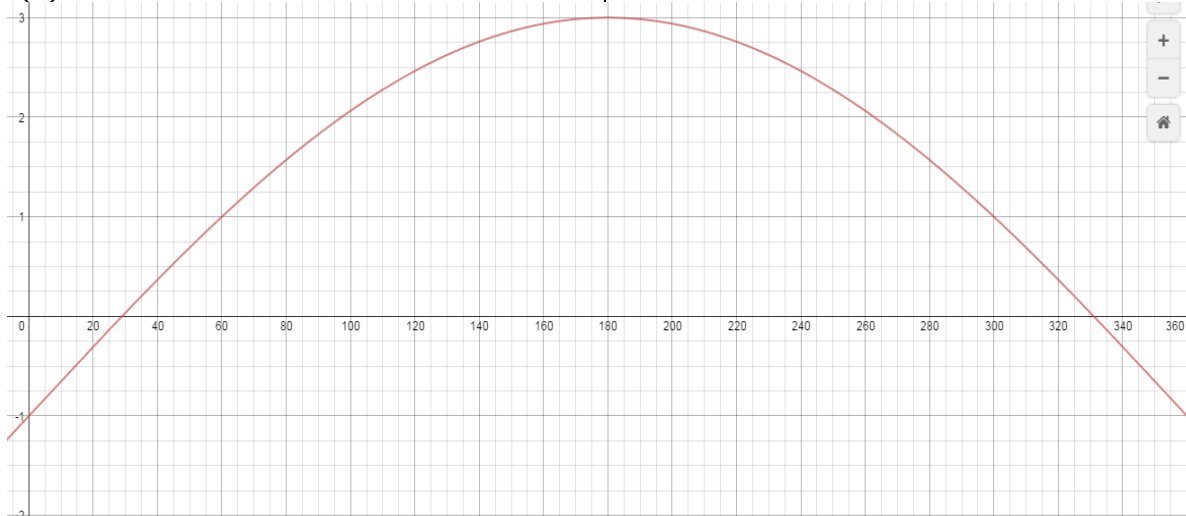
$= 2 + 2 \tan^2 x = 2 + \frac{2 \sin^2 x}{\cos^2 x}$

$\frac{2 \cos^2 x + 2 \sin^2 x}{\cos^2 x} = \frac{2(\cos^2 x + \sin^2 x)}{\cos^2 x}$

$= \frac{2}{\cos^2 x} = 2 \sec^2 x = RHS \quad \checkmark \checkmark \checkmark$

### Question 4

(a)



Period :  $720^\circ$

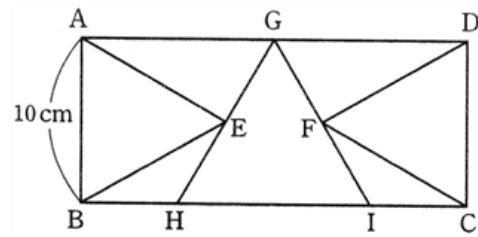
Range :  $-5 \leq y \leq 3$

Starts at  $(0^\circ, -1)$  ends at  $(360^\circ, -1)$

Maximum at  $(180^\circ, 3) \quad \checkmark \checkmark \checkmark$

(b)

i)



$\angle EAG = 90^\circ - 60^\circ = 30^\circ$

$\angle EGA = \angle GHI = 60^\circ$  (alternate angles on parallel lines)

$\therefore \angle AEG = 180^\circ - 60^\circ - 30^\circ$

(Angle sum of a triangle)

$= 90^\circ \quad \checkmark$

ii) AG = hypotenuse of AEG

$\cos 30^\circ = \frac{10}{AG}, \quad AG = \frac{20}{\sqrt{3}}$

$AD = 2 \times AG = \frac{40}{\sqrt{3}} \text{ cm} \quad \checkmark \checkmark$